Ram Prasad Bidiyasar

+91-8209641119 | ramprasad.mnit@gmail.com | LinkedIn | GitHub | LeetCode | Portfolio

Experience

Airtel Digital — Gurugram, India
Senior Software Engineer, Xstream backend — *Jul. 2025 – Present*

- Redesigned **Continue Watching** (resume where you left off) for the Xstream and IPTV apps. Watch events go through Kafka keyed by user ID, so writes from several devices stay in order, and sync latency fell by nearly 70%.
- Rebuilt watch-history reads (**Watch Info v2**) on Spring WebFlux and ScyllaDB: p99 latency down 75%, throughput per instance up about 4x.
- Built **multi-CDN routing and failover** for IPTV playback used by 10M+ active users. Rules stored in MongoDB pick the best CDN for each request, cutting buffering from 1% to 0.07%.
- Rebuilt the **event ingestion service** on Java 25: throughput per pod rose 6x (∼400 to ∼2,500 req/s) for 80–90K events/s at live-match peaks. If Kafka is down, events wait in Pub/Sub, so none are lost.
- Designed the IPTV **session service** on multi-zone Aerospike: one device per login, enforced atomically, with an SDK shared by the playback and SMS services.
- Raised personalized page generation from 30 to 100 TPS (about 30% lower latency), and added fallbacks so the Home screen stays up when a dependency fails.
- Stabilized and scaled **UCLM**, Airtel India's campaign messaging platform (SMS, email, WhatsApp; 20+ services, 1.5B+ messages a day), for its first migration load.
- Fixed production out-of-memory crashes and ScyllaDB failures, and cleared Black Duck and Coverity security findings across 20+ services.
- Contributed to an AI assistant that investigates playback and content issues, and wrote reusable agent rules and a partner-integration guide.

Airtel Digital — Gurugram, India
Software Engineer, Xstream backend — *Oct. 2023 – Jun. 2025*

- Migrated 20+ Xstream services from AWS to GCP with no downtime (SQS/Kinesis to Kafka, S3 to GCS, PostgreSQL, MongoDB, ScyllaDB); wrote most of the migration code.
- Cut content-enrichment delay from 1–2 days to near real time with a Debezium and Kafka pipeline from the Xstream catalog to data science.
- Upgraded the package and discovery services to Java 21 and Spring Boot 3.5, and fixed hourly 2–3 minute p99 spikes caused by log-file rollover.
- Onboarded Amazon Prime and Hotstar; built Prebid (ad bidding) analytics on Druid and a Keycloak login and roles module.

Projects

Aerospike Cluster Workspace | *Java 21, Spring Boot, React, Vite*

- Web tool to browse and edit records and check cluster health; long key-pattern jobs can be cancelled.

Reactive API Gateway | *Java 21, Spring Cloud Gateway, MongoDB, JWT, Docker*

- Gateway with MongoDB-backed dynamic routes, JWT auth, correlation IDs for tracing and admin APIs.

Technical Skills

Languages and frameworks: Java, C++, SQL, Spring Boot, Spring WebFlux, Vert.x, Kafka Streams
Data and messaging: Apache Kafka, Kafka Connect, ScyllaDB (Cassandra-compatible), Aerospike, PostgreSQL, MongoDB, Redis, Elasticsearch, Apache Druid, Google Pub/Sub
Cloud and infra: Google Cloud Platform (GCP), Amazon Web Services (AWS), Kubernetes, Docker, Jenkins, Apache Airflow
Practices: distributed systems, microservices, failover design, observability (Grafana, Kibana, AppDynamics)

Education

Malaviya National Institute of Technology Jaipur — Jaipur, India
B.Tech, Electronics and Communication Engineering (CGPA 8.16) — *July 2019 – May 2023*